

Punjab University BSCS (CC-311)

Operating Systems Notes by Syeda Laiba

Topic 1: Operating System Basics

Introduction

An **Operating System (OS)** is a crucial system software that acts as a bridge between the user and the computer hardware. It provides an environment where users can execute programs conveniently and efficiently. Without an OS, interacting directly with the hardware would be complex and nearly impossible for most users.

Think of the OS as the **manager of a company**: it assigns tasks to **employees** (hardware components), ensures work is done efficiently, resolves conflicts, and keeps everything running smoothly.

Components of a Computer System

A computer system can be viewed as comprising four main components:

1. **Hardware:** The physical components of the system, including the CPU, memory, and I/O devices.
2. **Operating System:** Controls and coordinates the use of hardware among different application programs and users.
3. **Application Programs:** Define how system resources are used to solve user problems (e.g., browsers, word processors, media players).
4. **Users:** People, machines, or other computers that utilize the system.

Together, these elements form a **cohesive** environment that transforms raw hardware into a useful computing system.

Role of the Operating System

The role of the OS can be understood from two perspectives:

1. User View:

- For **personal computers**, the OS focuses on ease of use and user experience.
- For **servers**, it emphasizes efficient resource management and fairness among users.
- For **mobile devices**, it manages touch interfaces, power consumption, and app performance.

2. System View:

From the hardware's viewpoint, the OS acts as a **resource manager**, responsible for: - Allocating CPU time, memory, disk space, and I/O devices. - Controlling system execution to prevent misuse or errors.

Goals of an Operating System

1. **Convenience:** Makes the computer easier to use.
 2. **Efficiency:** Utilizes system resources effectively.
 3. **Ability to Evolve:** Supports system updates, new features, and hardware changes.
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Functions of the Operating System

The OS performs multiple functions to ensure smooth and efficient system operation:

- **Process Management:** Handles creation, scheduling, and termination of processes. For example, when multiple apps run, the OS ensures they share the CPU fairly.
 - **Memory Management:** Allocates and deallocates memory space to programs as needed and keeps track of memory usage.
 - **Storage Management:** Manages file systems, directories, and disk space. It also controls disk scheduling and free-space allocation.
 - **I/O System Management:** Provides a consistent interface between hardware devices and user programs.
 - **Protection and Security:** Prevents unauthorized access to data and ensures system stability.
 - **User Interface (UI):** Provides the means for user interaction via CLI (Command Line Interface) or GUI (Graphical User Interface).
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Operating System as a Resource Manager

The OS allocates resources (CPU, memory, and storage) among programs and users. It aims to maintain: - **Efficiency:** Maximizing hardware utilization. - **Fairness:** Preventing any process from monopolizing system resources. - **Security:** Ensuring no process interferes with another.

Example: In a university computer lab, the OS ensures every student's process gets fair CPU time without crashing others' work.

Types of Operating Systems

Operating systems have evolved to meet diverse needs, from early batch systems to modern mobile and cloud platforms.

1. Batch Operating System

- **Description:** Oldest type; jobs are submitted in batches without direct user interaction.
- **Example:** IBM OS/360, Early UNIX.
- **Advantages:** Efficient for repetitive large-volume jobs.
- **Disadvantages:** No interactivity; debugging is difficult.

2. Multiprogramming Operating System

- **Description:** Multiple programs reside in memory simultaneously; OS switches between them to utilize the CPU efficiently.
- **Example:** UNIX, IBM OS/360.

3. Time-Sharing (Multitasking) Operating System

- **Description:** Extension of multiprogramming; CPU time is divided into small slices among users or tasks.
- **Example:** Windows, macOS, Linux.
- **Real-World Example:** Multiple users editing shared files on Google Docs simultaneously.

4. Distributed Operating System

- **Description:** Manages a group of networked computers as a single system.
- **Example:** LOCUS, Amoeba.
- **Advantage:** High reliability and resource sharing.

5. Network Operating System (NOS)

- **Description:** Manages network resources like file sharing and communication.
- **Example:** Windows Server, UNIX Network OS.

6. Real-Time Operating System (RTOS)

- **Description:** Used where timing is critical — responses must meet strict deadlines.
- **Examples:** VxWorks, QNX, RTLinux.
- **Use Case:** Airplane control systems, medical devices.

7. Embedded Operating System

- **Description:** Built into devices (e.g., routers, smartwatches) for specific functions.
- **Example:** Android (mobile), TinyOS, Embedded Linux.

8. Mobile Operating System

- **Description:** Specialized OS for smartphones and tablets.
- **Examples:** Android, iOS.

9. Multiprocessor Operating System

- **Description:** Designed for systems with multiple CPUs or cores.
- **Examples:** Linux SMP Kernel, Windows Server.

10. Cloud and Virtualization-Based Operating Systems

- **Description:** Manage virtual machines and resources in cloud environments.
- **Examples:** VMware ESXi, Microsoft Azure Hypervisor.

Examples of Popular Operating Systems

- **Windows:** General-purpose, GUI-based by Microsoft.
- **UNIX/Linux:** Multiuser, multitasking, open-source OS.
- **macOS:** UNIX-based hybrid OS with graphical interface.
- **Android:** Linux-based mobile OS.
- **iOS:** Derived from macOS for Apple devices.

Importance of an Operating System

Without an OS, users would have to manage every hardware operation manually — from memory access to I/O device control. The OS simplifies this by **abstracting hardware complexity**, offering a user-friendly and efficient computing experience.

Example: When you open a Word document, the OS manages memory allocation, keyboard input, display output, and storage access seamlessly in the background.

Quick Revision Cards

- **OS Definition:** Software that acts as an interface between user and hardware.
- **Main Components:** Hardware, OS, Application Programs, Users.
- **Key Roles:** Resource manager, control program.
- **Main Goals:** Convenience, efficiency, evolution.
- **Core Functions:** Process, memory, storage, I/O, security, user interface.
- **OS Types:** Batch, Multiprogramming, Time-Sharing, Distributed, Network, Real-Time, Embedded, Mobile, Multiprocessor, Cloud.
- **Example OS:** Windows, Linux, macOS, Android, iOS.
- **Importance:** Simplifies hardware use and enhances performance.

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